

MAJAVATN V, Norway

WMO: 011320

Lat: 65.1661N

Lon: 13.3667E

Elev: 319

StdP: 97.55

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.4	(c) -22.9	(d) -29.4	(e) 0.3	(f) -26.3	(g) -25.6	(h) 0.4	(i) -22.7	(j) 10.1	(k) -6.3	(l) 8.6	(m) -4.9	(n) 0.9	(o) 20	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	24.6	15.6	22.2	14.6	20.0	13.5	16.6	23.0	15.3	20.6	14.2	18.9	2.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	10.5	18.8	12.8	9.6	17.3	11.7	8.9	16.4	47.6	23.0	43.7	20.7	40.6	18.9	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.0	5.2	-28.0	27.0	3.1	2.6	-30.2	28.9	-32.0	30.4	-33.8	31.9	-36.1	33.7
			-28.2	17.3	3.1	2.0	-30.4	18.7	-32.2	19.9	-33.9	21.0	-36.2	22.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-7.0	-6.8	-4.4	0.8	5.2	10.0	13.8	12.1	8.6	2.8	-1.7	-4.8		
	DBStd	8.61	6.74	6.89	5.69	3.50	3.53	3.45	3.62	2.80	3.08	3.40	5.42	6.26		
	HDD10.0	3031	526	471	447	277	157	42	6	11	61	225	351	459		
	HDD18.3	5813	784	704	706	527	407	252	147	194	292	483	601	718		
	CDD10.0	268	0	0	0	0	8	41	124	76	19	1	0	0		
	CDD18.3	8	0	0	0	0	0	1	7	1	0	0	0	0		
	CDH23.3	126	0	0	0	0	3	14	102	6	0	0	0	0		
	CDH26.7	22	0	0	0	0	0	1	21	0	0	0	0	0		
(14) Wind	WSAvg	2.1	2.5	2.6	2.4	2.0	1.9	1.8	1.7	1.8	2.0	2.2	2.2	2.5		
(15) Precipitation	PrecAvg	1397	147	143	141	86	63	79	86	88	137	145	125	158		
	PrecMax	1870	328	340	293	211	111	173	216	165	381	290	278	307		
	PrecMin	1007	11	9	22	25	17	16	39	19	41	50	7	14		
	PrecStd	218	86	86	78	48	21	36	39	32	77	62	75	87		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	4.9	5.8	13.7	21.9	25.2	29.4	24.2	19.6	13.0	8.7	5.4		
		MCWB	2.9	3.2	2.3	6.9	12.3	14.6	18.4	15.7	12.9	10.1	7.4	4.2		
	2%	DB	3.1	3.7	4.5	10.5	17.9	22.0	26.3	22.1	16.9	10.5	6.9	4.2		
		MCWB	1.9	2.4	2.1	5.3	10.5	13.2	16.6	14.9	11.8	7.7	5.4	2.8		
	5%	DB	2.0	2.9	3.5	7.7	14.8	19.1	23.6	20.1	15.1	9.0	5.0	3.1		
		MCWB	0.7	1.6	1.7	3.6	8.7	12.1	15.8	14.0	11.3	6.7	3.4	1.6		
	10%	DB	1.0	1.7	2.6	5.8	11.8	16.7	21.2	18.0	13.6	7.4	4.0	2.2		
		MCWB	0.0	0.3	1.0	2.5	7.3	11.0	14.5	12.9	10.2	5.6	2.5	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.5	3.5	7.2	13.0	15.8	19.9	17.3	13.7	9.9	7.5	4.5		
		MCDB	4.3	4.4	4.8	13.3	20.6	23.0	28.0	22.2	17.2	12.4	8.5	5.2		
	2%	WB	2.1	2.6	2.6	5.5	11.2	14.1	17.7	15.8	12.6	8.0	5.4	3.0		
		MCDB	2.9	3.7	4.0	10.0	16.5	20.2	24.2	20.2	16.1	10.0	6.8	3.9		
	5%	WB	0.9	1.7	1.8	4.0	9.1	12.9	16.2	14.5	11.5	6.9	3.6	1.7		
		MCDB	1.6	2.8	3.3	6.9	14.4	18.1	22.4	18.8	14.8	8.7	4.6	2.7		
	10%	WB	0.0	0.4	1.0	3.0	7.3	11.5	14.9	13.5	10.5	5.8	2.6	0.9		
		MCDB	1.0	1.5	2.5	5.4	11.4	16.0	20.2	17.7	13.1	7.4	3.8	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.4	9.0	8.5	8.7	8.5	9.3	9.2	7.2	5.5	4.7	6.1		
		MCDBR	4.9	4.7	6.5	11.2	14.7	14.3	14.3	13.5	10.5	7.2	4.2	5.0		
		MCWBR	4.6	4.0	4.9	6.7	8.1	6.8	6.3	6.9	6.3	5.1	3.6	4.4		
		MCWBR	4.8	4.9	5.5	9.9	13.8	12.4	12.9	11.6	9.4	6.1	4.2	5.0		
(40) Clear-Sky Solar Irradiance	taud	taub	0.194	0.265	0.297	0.326	0.333	0.333	0.342	0.329	0.309	0.285	0.353	0.198		
		taud	1.923	2.142	2.183	2.219	2.348	2.472	2.478	2.503	2.546	2.379	2.702	1.935		
		Ebn at Noon	305	618	771	839	872	882	861	840	782	611	298	71		
		Edh at Noon	22	57	87	107	104	95	92	81	63	45	21	7		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.10	0.51	1.80	3.54	4.53	4.72	4.47	3.48	1.86	0.76	0.17	0.03		
		RadStd	0.01	0.10	0.29	0.38	0.33	0.58	0.55	0.39	0.25	0.12	0.03	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page